SYNOPSIS

Report on

UCare: Platform for Caregivers

by

Tanishka Gupta 202410116100218
Shipra Upadhyay 202410116100196
Sonam Dobriyal 202410116100210

Session:2025-2026 (IV Semester)

Under the supervision of
Dr. Neelam Rawat
Ms. Shruti Aggarwal

KIET Group of Institutions, Delhi-NCR, Ghaziabad

DEPARTMENT OF COMPUTER APPLICATIONS
KIET GROUP OF INSTITUTIONS, DELHI-NCR,
GHAZIABAD-201206
(2025 - 2026)

ABSTRACT

This project focuses on the development of **UCare**, a digital platform designed to provide emotional and psychological support to individuals whose family members are involved in addiction or excessive alcohol consumption. Addiction not only affects the individual but also deeply impacts family members, leading to stress, anxiety, and mental health challenges.

Unlike traditional counseling systems that require physical visits and professional assistance, the proposed platform leverages **Artificial Intelligence and web-based technologies** to provide real-time mental health support through an AI-powered chatbot and an online community support system. The chatbot interacts with users, understands their emotional state, and provides guidance, coping strategies, and awareness about addiction-related issues. Additionally, the community feature allows users to share their experiences anonymously and receive peer support from others facing similar challenges.

The platform is designed to be accessible, user-friendly, and scalable, enabling users to receive support anytime and anywhere through the internet. UCare aims to reduce emotional stress, increase awareness, and create a safe digital space for affected family members.

Key highlights of the project include:

- **AI-Based Mental Health Chatbot:** Provides emotional support, guidance, and coping strategies through conversational interaction.
- **Community Support System:** Enables anonymous discussion and peer-to-peer support among users.
- **Accessibility & Scalability**: Requires only a web-enabled device and internet connection, making it easily accessible to a wide audience.
- **Mental Health Awareness & Education:** Provides resources and information about addiction and its impact on families.
- **Research Contribution**: Demonstrates the application of AI and digital platforms in mental health support and social well-being.

TABLE OF CONTENTS

INTRODUCTION

Addiction and drinking problems are serious issues that affect not only the addicted person but also their family members. Family members often feel stressed, sad, scared, and embarrassed, which affects their mental health. Many people cannot get help from doctors or counselors because it is costly, far away, or they feel shy to talk about it. With the help of Artificial Intelligence and the internet, support can be given online. **UCare** is a digital platform that provides help through a chatbot and a community feature so people can share their problems and get support safely.

Key Points:

1. Need for Mental Health Support

- Addiction impacts not only individuals but also their family members emotionally and psychologically.
- Family members often face stress, anxiety, and social isolation.

2. Limitations of Traditional Support Systems

- Counseling and therapy sessions are costly and not easily accessible.
- Social stigma prevents people from seeking help.
- Physical support groups are location and time dependent.

3. Role of Technology

- AI and web technologies enable real-time, scalable, and anonymous support.
- Chatbots and online communities can provide immediate emotional assistance.

4. Purpose of UCare Platform

- To provide emotional guidance using an AI-based chatbot.
- To create a community platform for peer support and experience sharing.
- To increase awareness about addiction and its impact on families.

5. Social and Academic Significance

- Helps improve mental well-being of affected individuals.
- Demonstrates the application of AI in mental health and social welfare domains.
- Contributes to research in digital mental health support systems.

REVIEW OF EXISTING SYSTEM

The field of digital mental health and AI-assisted emotional support has gained significant attention in recent years, particularly in areas such as online counseling, mental wellness applications, and AI-powered chatbots. Researchers have explored Natural Language Processing (NLP), sentiment analysis, and machine learning models to provide real-time emotional assistance. While many platforms focus on general mental health, the specific support for family members of addicted individuals remains relatively underexplored.

Key Points

1. Traditional Mental Health Support Approaches

Historically, mental health support has been provided through in-person counseling, therapy sessions, and support groups.

Challenges include:

- **Accessibility Issues**: Limited availability in rural or remote areas.
- **Cost Factor**: Therapy and counseling sessions can be expensive.
- **Social Stigma**: Many individuals hesitate to seek help openly.
- **Time Constraints**: Fixed appointments may not suit everyone's schedule.
- **Limited Focus**: Most programs prioritize addicted individuals rather than affected family members.

2. Early Applications of AI in Mental Health

Several AI-based mental health applications have emerged, such as chatbot-based support systems.

Examples include:

- AI Chatbots for stress and anxiety management.
- Mood tracking applications using sentiment analysis.
- Digital cognitive behavioral therapy (CBT) platforms.

These systems demonstrate that AI-driven conversational agents can provide 24/7 emotional assistance and encourage users to share feelings anonymously.

3. Natural Language Processing (NLP) Frameworks

The core technology enabling AI chatbots is NLP.

Popular technologies include:

- **BERT and GPT models** for conversational understanding.
- **Sentiment Analysis models** for detecting emotional state.
- **Dialogflow and Rasa** for chatbot development.

These frameworks are widely used in mental health bots, customer service bots, and virtual assistants.

4. Online Community Support Systems

Research highlights the importance of peer-support communities in improving emotional well-being.

- Anonymous discussion forums reduce stigma.
- Peer interaction increases emotional resilience.
- Shared experiences help reduce loneliness.

However, most online communities are not specifically designed for families of addicted individuals.

5. Research Gap and Opportunity

Existing literature shows strong development in AI-based mental health tools and community platforms, but there is limited focus on integrated systems specifically supporting family members of addicted individuals.

This project fills the gap by:

- Providing an AI-powered mental health chatbot tailored to affected family members.
- Integrating a secure and anonymous community support feature.
- Delivering a scalable web/mobile platform accessible anytime, anywhere.

OBJECTIVE

The primary objective of the UCare project is to design and develop an AI-driven digital platform that provides emotional and psychological support to individuals whose family members are affected by addiction or excessive alcohol consumption. The platform aims to bridge the gap between mental health needs and accessible digital solutions by integrating chatbot assistance with community support features.

Specific Research Objectives

1. **To analyze the psychological impact of addiction on family members**
   Study how addiction affects emotional well-being, stress levels, anxiety, and social behavior of family members.
2. **To design an AI-powered mental health chatbot**
   Develop a conversational chatbot using Natural Language Processing (NLP) that can provide emotional support, coping strategies, and basic guidance.
3. **To implement sentiment analysis techniques**
   Detect user emotions (such as stress, sadness, anger) and respond with appropriate supportive messages.
4. **To create a secure and anonymous community support system**
   Provide a safe digital space where users can share experiences and connect with others facing similar challenges.
5. **To ensure accessibility and scalability**
   Design the platform as a web/mobile-based solution that can be accessed anytime, anywhere.
6. **To evaluate system effectiveness**
   Measure user satisfaction, emotional improvement, and usability of the platform through feedback and testing.
7. **To promote mental health awareness**
   Encourage open conversations and reduce stigma related to addiction and mental health issues.

HARDWARE AND SOFTWARE REQUIREMENTS

For the successful design and implementation of the UCare – AI-Based Mental Health & Community Support Platform, both hardware and software resources are essential. The requirements are kept minimal to ensure accessibility and scalability, allowing users to access emotional support services using commonly available devices such as laptops, smartphones, or desktops.

Hardware Requirements:

- Device: Webcam-enabled Laptop / Desktop / Smartphone
- Processor: Intel i5 / AMD Ryzen 5 or above
- RAM: Minimum 8 GB (Recommended for smooth development and AI processing)
- Storage: 256 GB SSD or higher
- Internet Connection: Stable broadband or 4G/5G for real-time chatbot interaction and community synchronization

Software Requirements:

- **Programming Languages:**
  Python 3.10+ for robust backend logic and AI integration
- **Frontend:**
  React for building dynamic and responsive user interfaces
- **Backend:**
  Flask, a lightweight and powerful micro-framework for API development
- **Database:**
  Firebase for real-time data storage and synchronization
- **AI/ML Frameworks:**
  RAG (Retrieval Augmented Generation) and LangChain for advanced chatbot capabilities
- **Tools/IDE:**
  Visual Studio Code as our primary Integrated Development Environment

METHODOLOGY

The research methodology for the UCare platform focuses on designing, developing, and evaluating an AI-driven mental health support system specifically for individuals whose family members are affected by addiction. The methodology follows a structured and systematic approach to ensure reliability, usability, and effectiveness of the proposed system.

1. Problem Identification and Requirement Analysis

The first step involves identifying the psychological challenges faced by family members of addicted individuals. Literature review and surveys are conducted to understand emotional stress, anxiety levels, and lack of accessible support systems. Based on the findings, system requirements such as chatbot support, sentiment analysis, and community interaction are defined.

2. System Design

A modular system architecture is designed consisting of:

- User Interface (React-based frontend)
- Backend API (Flask framework)
- AI Chatbot Module (RAG + LangChain integration)
- Database (Firebase for real-time storage)
- Community Support Module
- The design ensures scalability, security, and real-time interaction.

3. AI Model Development

The chatbot is developed using:

- Natural Language Processing (NLP) techniques
- Retrieval-Augmented Generation (RAG) for context-aware responses
- Sentiment analysis to detect user emotions
- LangChain is integrated to manage conversational flow and memory.

4. Implementation

Frontend and backend modules are developed and integrated through REST APIs. Firebase is configured for authentication and real-time community discussions. The chatbot is trained and tested using sample emotional conversation datasets.

5. Testing and Evaluation

- Functional Testing: Ensures all modules work correctly
- Usability Testing: Collects user feedback
- Performance Testing: Evaluates response time
- Security Testing: Ensures data privacy and confidentiality

6. Result Analysis

User feedback and system performance metrics are analyzed to measure effectiveness in providing emotional support and improving user engagement.

Conclusion

This research methodology ensures a structured development process, combining AI technology with mental health support principles to deliver a reliable, scalable, and user-friendly digital platform.

OUTCOME

The development and implementation of the **UCare platform** resulted in a functional AI-driven mental health support system designed specifically for individuals whose family members are affected by addiction or alcohol dependency. The research successfully demonstrated how Artificial Intelligence and digital community support can be integrated to provide accessible, anonymous, and real-time emotional assistance.

The platform effectively combines an AI-powered chatbot with a secure community support feature, creating a safe digital environment where users can share their emotions without fear of judgment or stigma.

Key Outcomes

- **Successful Development of AI Chatbot**
  The chatbot provides real-time emotional support using NLP, RAG, and LangChain integration.
- **Emotion Detection Capability**
  Sentiment analysis helps identify user emotional states such as stress, sadness, or anxiety and responds accordingly.
- **Community Support System Implementation**
  Users can connect anonymously, share experiences, and receive peer encouragement.
- **Improved Accessibility**
  The web-based system ensures 24/7 availability without geographical limitations.
- **Reduced Social Stigma**
  Anonymous interaction encourages users to express feelings openly.
- **Scalable Architecture**
  Firebase-based real-time database allows the platform to handle multiple users efficiently.
- **Positive User Feedback (Expected/Observed)**
  Initial testing indicates improved emotional comfort and user engagement.

PROPOSED TIME DURATION

Phase	Duration
Literature Review	Week 1 – 2
Data Collection	Week 3
Translation & Preprocessing	Week 4
Model Training	Week 5-6
Model Testing	Week 7
Result Analysis & Visualization	Week 8-9
Report & Presentation	Week 10

REFERENCES

- World Health Organization (WHO). (2023). *Global status report on alcohol and health.* Geneva: WHO Press.
- National Institute on Drug Abuse (NIDA). (2022). *Substance use and family impact research findings.* U.S. Department of Health and Human Services.
- Fitzpatrick, K. K., Darcy, A., & Vierhile, M. (2017). Delivering cognitive behavioral therapy to young adults with symptoms of depression and anxiety using a fully automated conversational agent (Woebot). *JMIR Mental Health*, 4(2), e19.
- Inkster, B., Sarda, S., & Subramanian, V. (2018). An empathy-driven, conversational artificial intelligence agent (Wysa) for digital mental well-being. *JMIR mHealth and uHealth*, 6(11), e12106.
- Miner, A. S., Milstein, A., & Hancock, J. T. (2017). Talking to machines about personal mental health problems. *JAMA*, 318(13), 1217–1218.
- Devlin, J., Chang, M. W., Lee, K., & Toutanova, K. (2019). BERT: Pre-training of deep bidirectional transformers for language understanding. *NAACL-HLT Conference Proceedings.*
- Lewis, P., Perez, E., Piktus, A., et al. (2020). Retrieval-Augmented Generation for knowledge-intensive NLP tasks. *Advances in Neural Information Processing Systems (NeurIPS).*
- Firebase. (2023). *Firebase Documentation.* Google Developers.
- LangChain Documentation. (2023). *LangChain Framework for LLM Applications.*
- American Psychological Association (APA). (2020). *Publication manual of the American Psychological Association* (7th ed.).